Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Benzene (C6H6)
Basis set   : cc-pVDZ
Charge   : 0
Multiplicity : 1
Total e-   : 42
n-body order : 5

GROUND STATE ENERGIES

E0 (Ha) : -230.75355805
E1 (Ha) : -230.57826157
Gap (Ha) : 0.17529648
HF (Ha) : -230.71878874

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 8

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,6)

Frozen core orbs : 18
Frozen e-   : 36
Active e-   : 6
Active orbs   : 6
Spin-orbitals   : 12
Hilbert dim   : 400

SPIN PROPERTIES

<S2>   : 0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 1.503200
<D> / site : 0.250533

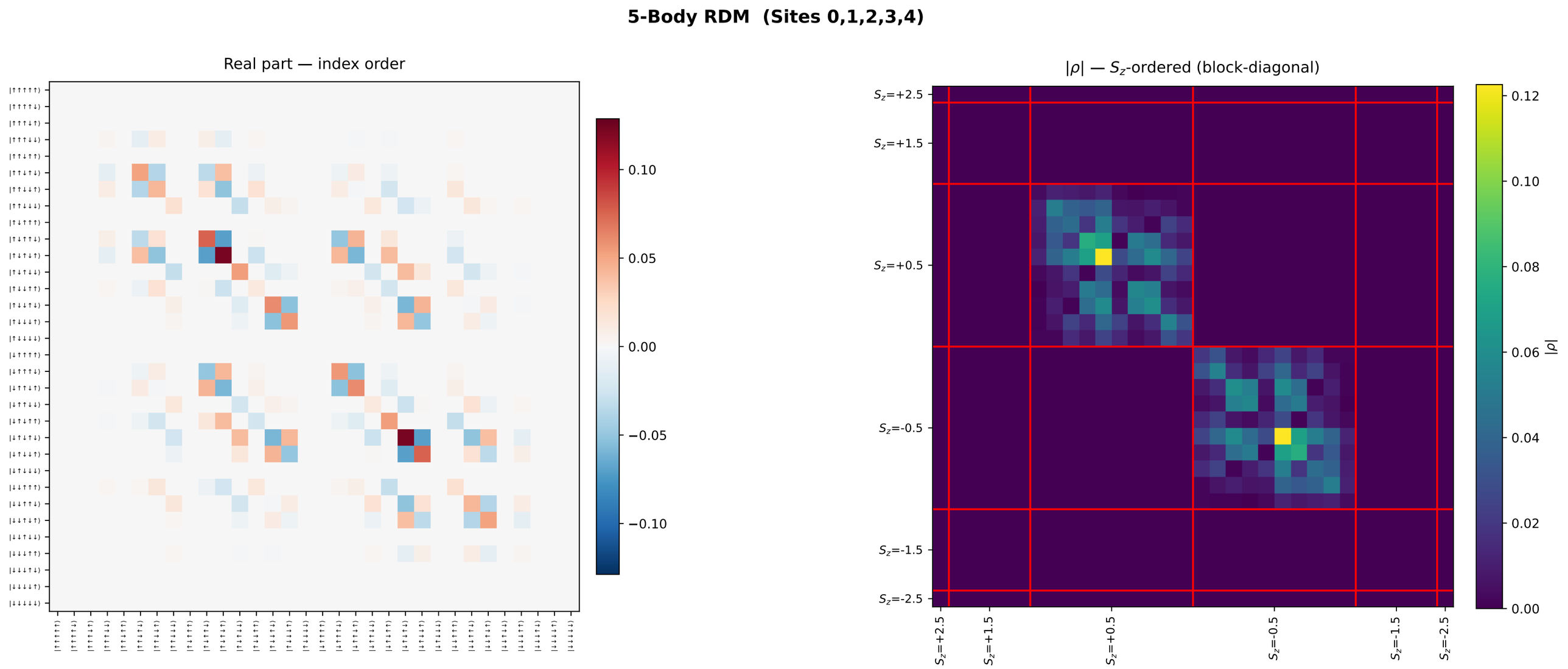
THIS PDF

Body order : 5
Clusters   : 6

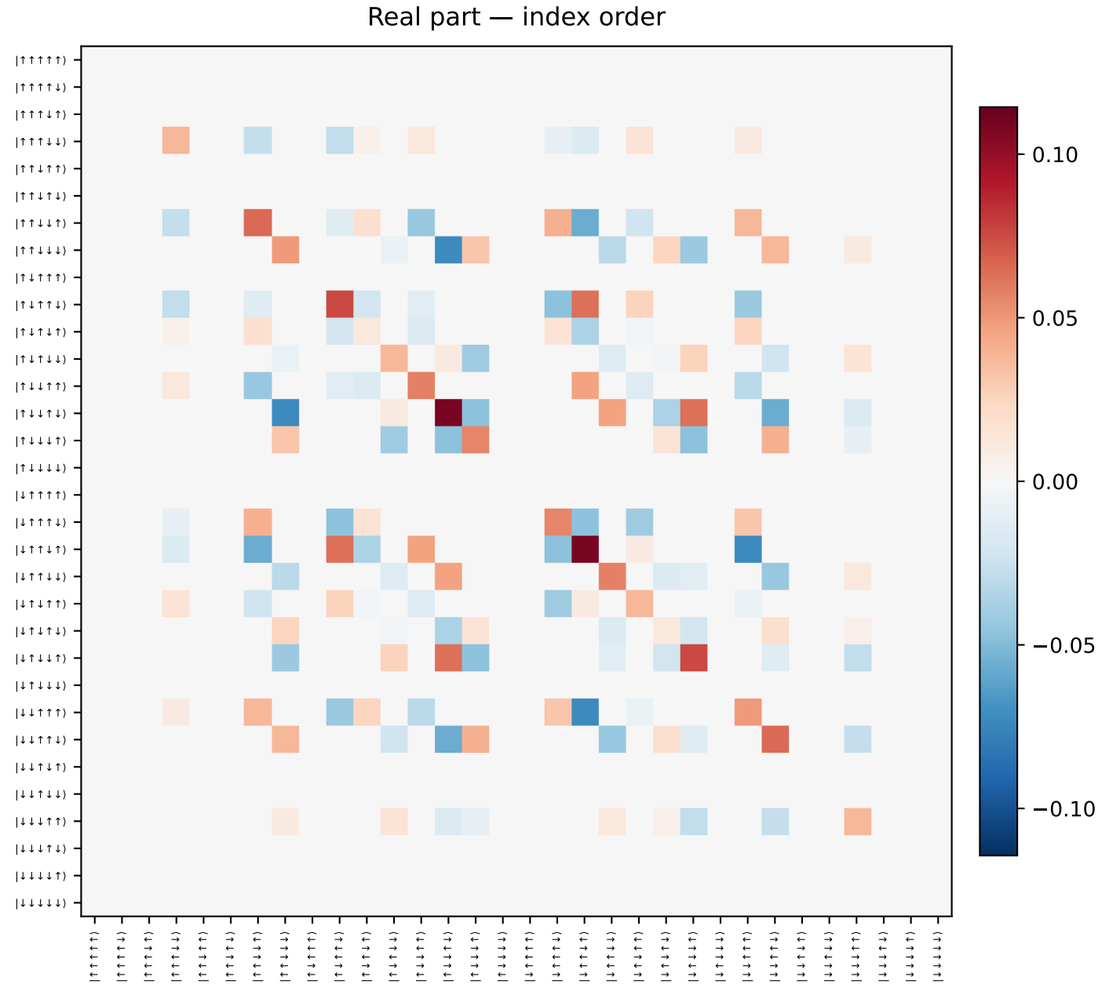
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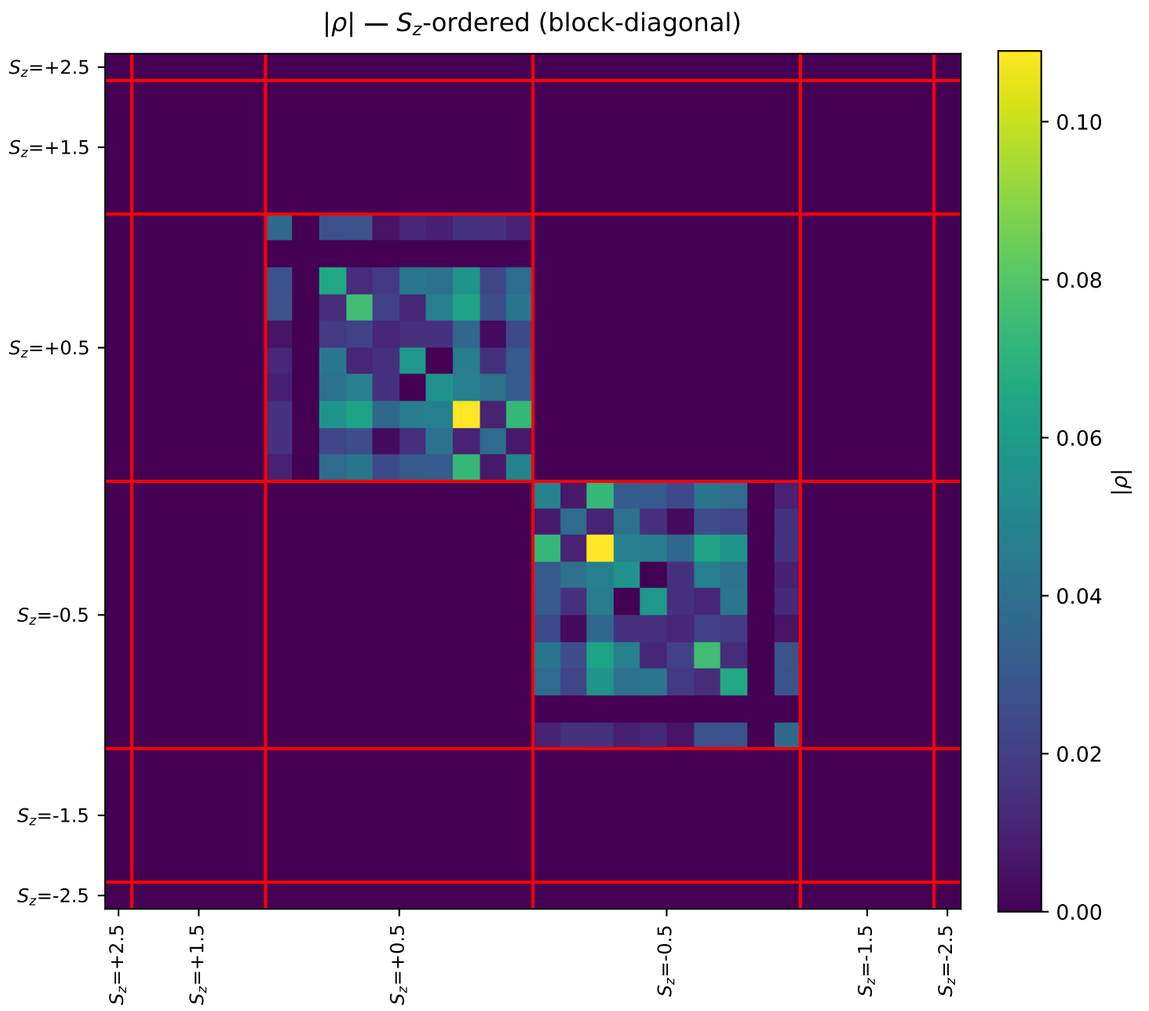
Summary — 5-Body RDMs (Benzene, CAS(6,6))

Orbitals	Purity	Entropy	C_GME (pure)	C_LB	C_gcg	C_UB	State
0,1,2,3,4	0.198754	1.726075	—	—	—	—	STRONGLY MIXED
0,1,2,3,5	0.214903	1.657210	—	—	—	—	MIXED
0,1,2,4,5	0.196300	1.732465	—	—	—	—	STRONGLY MIXED
0,1,3,4,5	0.221225	1.640947	—	—	—	—	MIXED
0,2,3,4,5	0.197867	1.729206	—	—	—	—	STRONGLY MIXED
1,2,3,4,5	0.216521	1.653706	—	—	—	—	MIXED

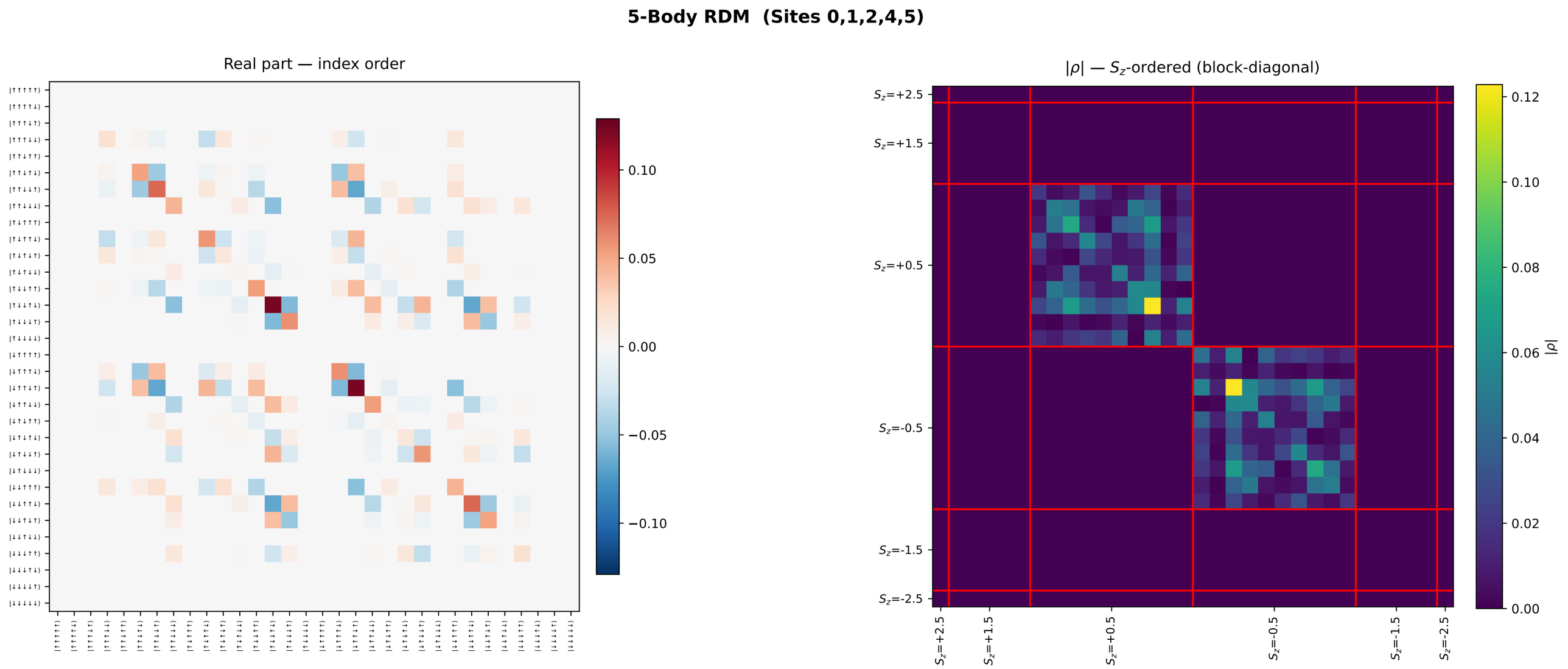

5-Body RDM (Sites 0,1,2,3,4)
Real part — index order
0.10
0.05
0.00
−0.05
−0.10
|ρ| — S_z-ordered (block-diagonal)
S_z=+2.5
S_z=+1.5
S_z=+0.5
S_z=-0.5
S_z=-1.5
S_z=-2.5
0.12
0.10
0.08
0.06
0.04
0.02
0.00
|ρ|
Trace=1.000000 Purity=0.198754 Entropy=1.726075 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

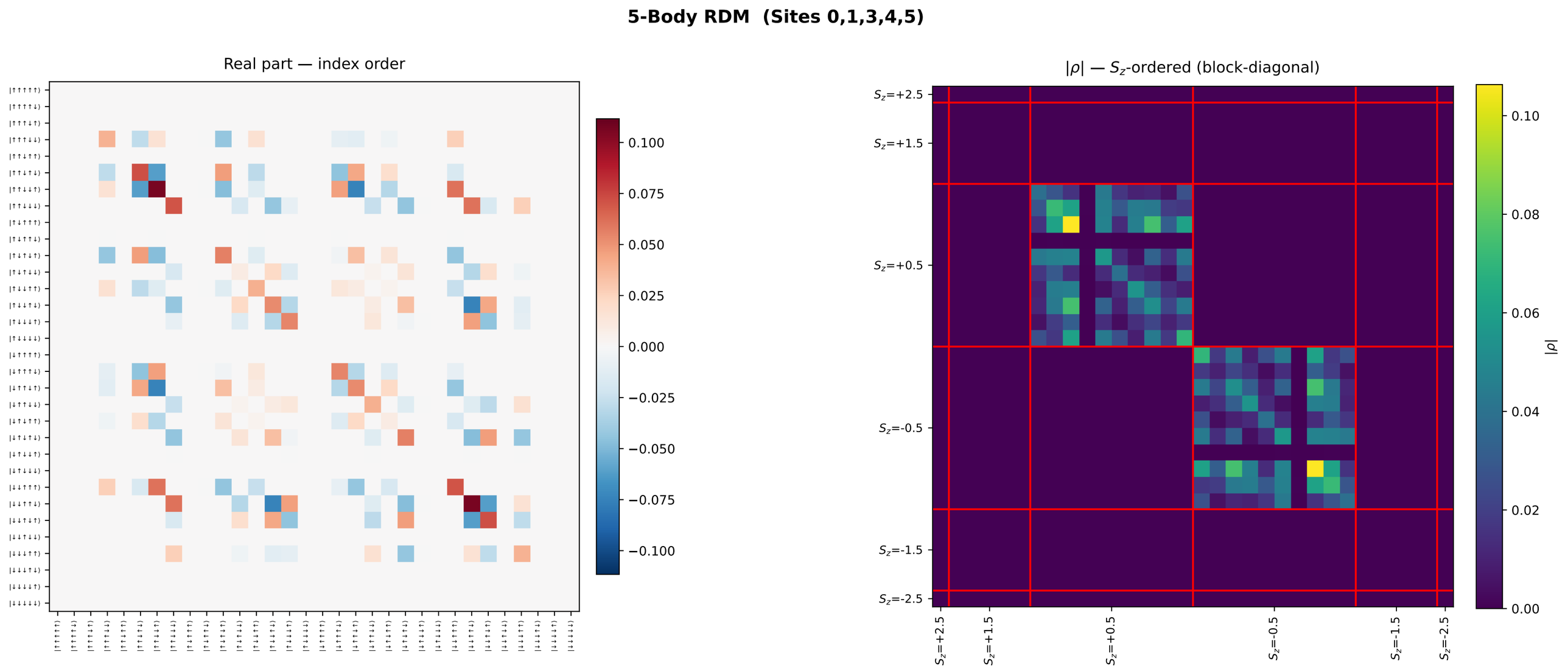
5-Body RDM (Sites 0,1,2,3,5)

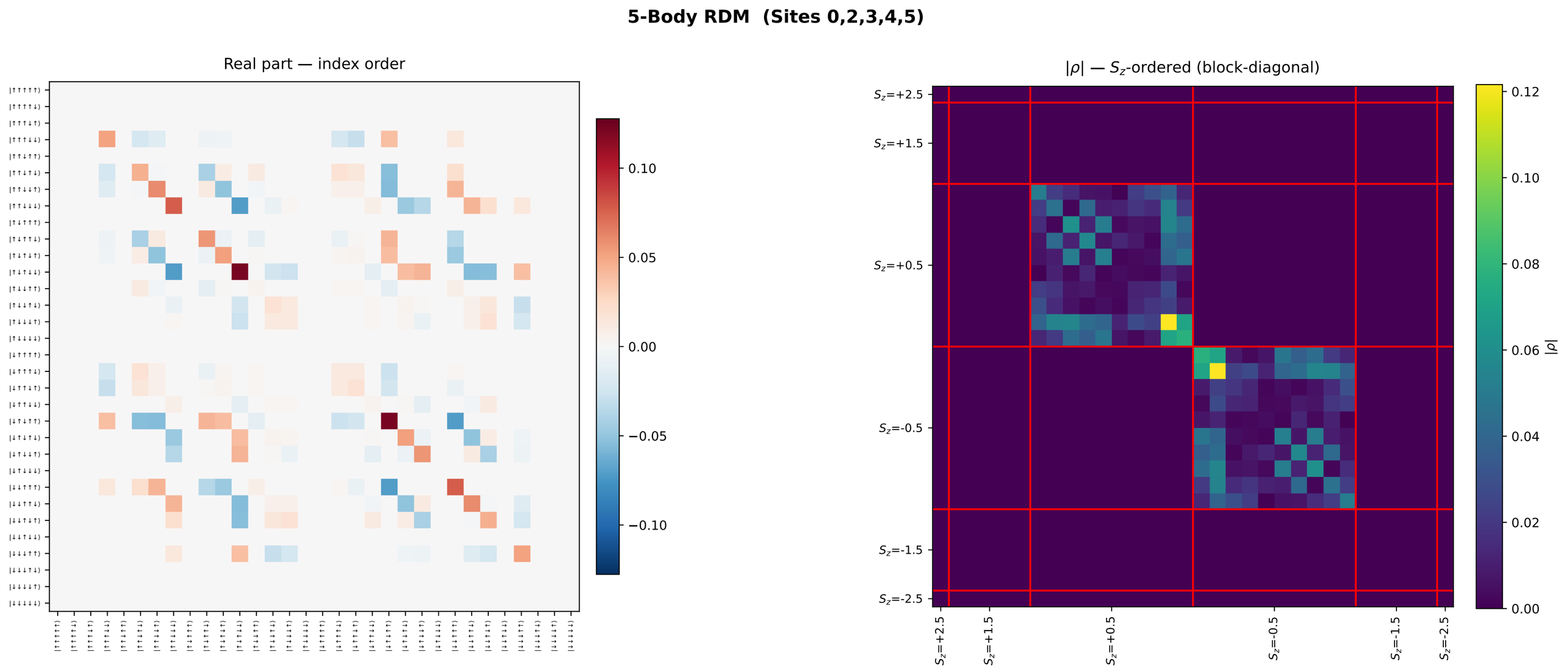


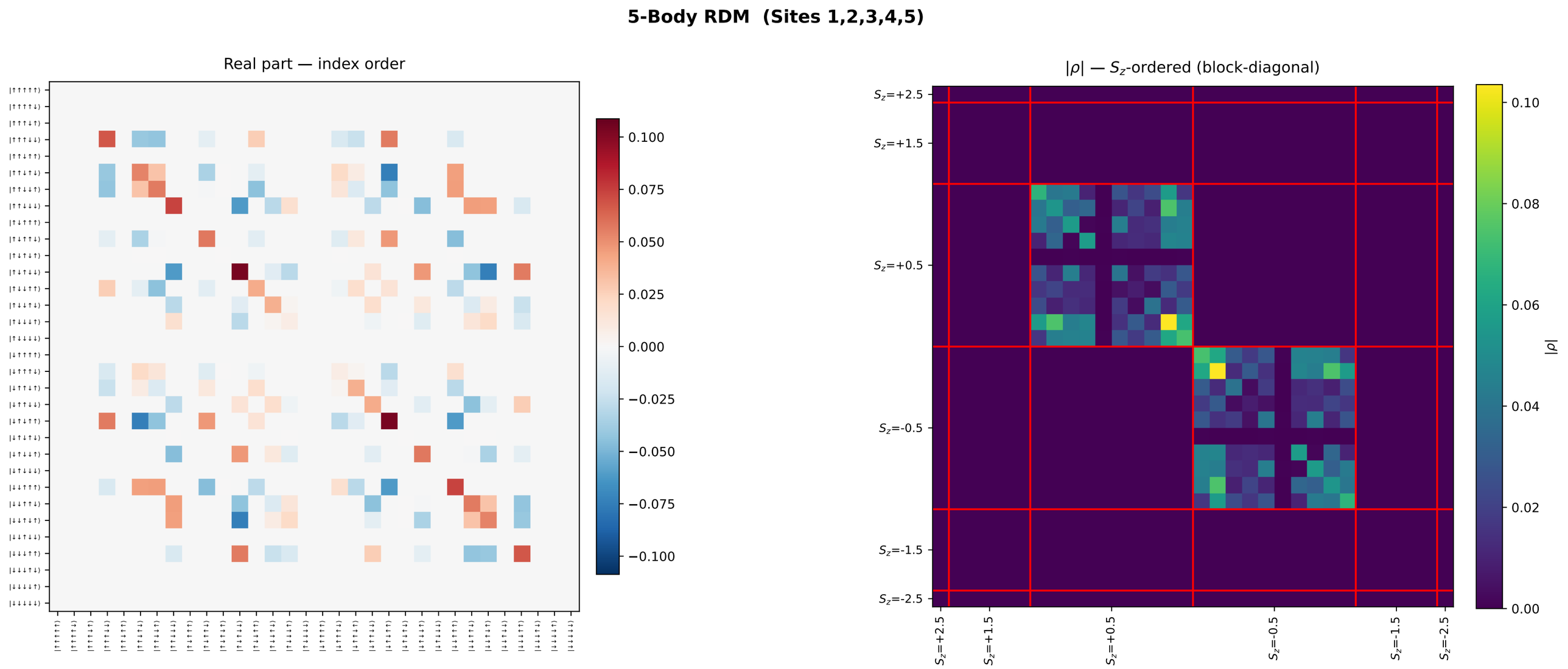


Trace=1.000000 Purity=0.214903 Entropy=1.657210 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)


5-Body RDM (Sites 0,1,2,4,5)
Real part — index order
|↑↑↑↑↑⟩
|↑↑↑↑↓⟩
|↑↑↑↓↑⟩
|↑↑↑↓↓⟩
|↑↑↓↑↑⟩
|↑↑↓↑↓⟩
|↑↑↓↓↑⟩
|↑↑↓↓↓⟩
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|↓↓↓↑↑⟩
|↓↓↓↑↓⟩
|↓↓↓↓↑⟩
|↓↓↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
|ρ| — S_z-ordered (block-diagonal)
S_z=+2.5
S_z=+1.5
S_z=+0.5
S_z=-0.5
S_z=-1.5
S_z=-2.5
0.12
0.10
0.08
0.06
0.04
0.02
0.00
|ρ|
Trace=1.000000 Purity=0.196300 Entropy=1.732465 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)


5-Body RDM (Sites 0,1,3,4,5)
Real part — index order
|↑↑↑↑↑⟩
|↑↑↑↑↓⟩
|↑↑↑↓↑⟩
|↑↑↑↓↓⟩
|↑↑↓↑↑⟩
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|↓↓↓↑↓⟩
|↓↓↓↓↑⟩
|↓↓↓↓↓⟩
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
|ρ| — S_z-ordered (block-diagonal)
S_z=+2.5
S_z=+1.5
S_z=+0.5
S_z=-0.5
S_z=-1.5
S_z=-2.5
|ρ|
0.10
0.08
0.06
0.04
0.02
0.00
Trace=1.000000 Purity=0.221225 Entropy=1.640947 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)


5-Body RDM (Sites 0,2,3,4,5)
Real part — index order
0.10
0.05
0.00
−0.05
−0.10
|ρ| — S_z-ordered (block-diagonal)
S_z=+2.5
S_z=+1.5
S_z=+0.5
S_z=-0.5
S_z=-1.5
S_z=-2.5
0.12
0.10
0.08
0.06
0.04
0.02
0.00
|ρ|
Trace=1.000000 Purity=0.197867 Entropy=1.729206 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)


5-Body RDM (Sites 1,2,3,4,5)
Real part — index order
|↑↑↑↑↑⟩
|↑↑↑↑↓⟩
|↑↑↑↓↑⟩
|↑↑↑↓↓⟩
|↑↑↓↑↑⟩
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|↓↓↓↑↓⟩
|↓↓↓↓↑⟩
|↓↓↓↓↓⟩
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
|ρ| — S_z-ordered (block-diagonal)
S_z=+2.5
S_z=+1.5
S_z=+0.5
S_z=-0.5
S_z=-1.5
S_z=-2.5
|ρ|
0.10
0.08
0.06
0.04
0.02
0.00
Trace=1.000000 Purity=0.216521 Entropy=1.653706 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)